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Studies on analysis of hormonal effects in regulating the blood volume in a freshwater teleostean fish *Mystus bleekeri* (Day, 1877)

Akash Garain* & Rajendra Mistry

University Department of Zoology, V.B. University, Hazaribag, Jharkhand, India

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Abstract- The effect of hydrocortisone acetate and L-thyroxine on blood volume and haematocrit values in *Mystus bleekeri* of almost same age group (23 - 35 grams) was studied at 28°C. The blood volume was determined by dye dilution technique using Evans blue. Different doses (total viable dose 1 and 3 mg/ml/100 g of hydrocortisone acetate and 0.5 and 1 mg/ml/100 g of L-thyroxine) of hormones were injected intraperitoneally in a period of 10 days (daily) in different sets of animals. Saline injected control animals had a total blood volume of 0.378 ± 0.051 ml, relative (per 100 g body weight) blood volume of 1.643 ± 0.221 ml and haematocrit value of 45.0 ± 1.22%. Hydrocortisone dose of 1 mg/100 had TBV of 0.278 ± 0.028 ml. RBV of 1 108 ± 0.113 ml and Ht value of 44.0 ± 0.96% Hydrocortisone dose of 3.0 mg/ml/100g had TBV of 0.222 ± 0.021 ml, RBV of 0.965 ± 0.090 ml and Ht value of 33.0 + 0.84". L-thyroxine treated animals (0.5 mg/ 100 g) had TBV of 0.417 ± 0.02 ml, RBV of 1.737 ± 0.125 ml and Ht value of 25.0 ± 0.75% L-thyroxine dose of 10W/100 had TBV or 0.253 0.026 ml, RBV of 1.100 = 0.117 ml and Tilt value of 28.0 ± 0.71%. From t-test of the data the value of "P" shows that in case of hydrocortisone injected animals the total and relative blood volume as well as haematocrit values become less in both the doses but it is more significant (P<0.001, 0.05 and 0.001) in higher dose. In case of L-thyroxin injected animals the results were somewhat different in different doses. In smaller dose total and relative blood volumes were increased (N S.) but the haematocrit value was decreased, whereas in higher dose blood volumes and the value of haematocrit were significantly decreased (P<0.005, 0.005 and 0.001). From the experiment it is inferred that both the hormones reduce blood volume and the haematocrit values in this air breathing fish (*Mystus bleekeri*).

Key words: Water quality index, river systems, Jharkhand state

INTRODUCTION

The blood volume estimations are of much importance in ascertaining the physiological conditions associated with loss or gain of fluid by the body. Several environmental, physiological and pathological conditions such as acute exposure to cold, high temperatures, age, muscular exercise, emotional excitement, pregnancy, hemorrhage, burns, dehydration, pernicious anaemia, obesity, myxoedema, polycythaemia vera, cirrhosis of the liver, leukemia, splenomegaly and congestive heart failure are responsible for changes in the blood volume of human being.[1] The neural lobe (neurohypophysis) of pituitary[1], thyroid[2], parathyroid and adrenal[1] have either direct or indirect effect on the maintenance of ionic balance which ultimately influences total body water content and blood volume.

Thorson (1958, 1959, 1961)[3-5] measured the body fluid compartments of various fishes and discussed its ecological implications in aquatic vertebrates. Prosser and

*Corresponding author :
Phone : 7209201692
E-mail : akashgarain@gmail.com

Weinstein (1950)[6] compared the blood volume in animals with open and closed circulatory systems. Houston and DeWilde (1969)[7] determined the effect of environmental temperature on the body fluid system of freshwater teleost. But the effect of hormones on blood volume and body fluid system in aquatic vertebrates is little known.

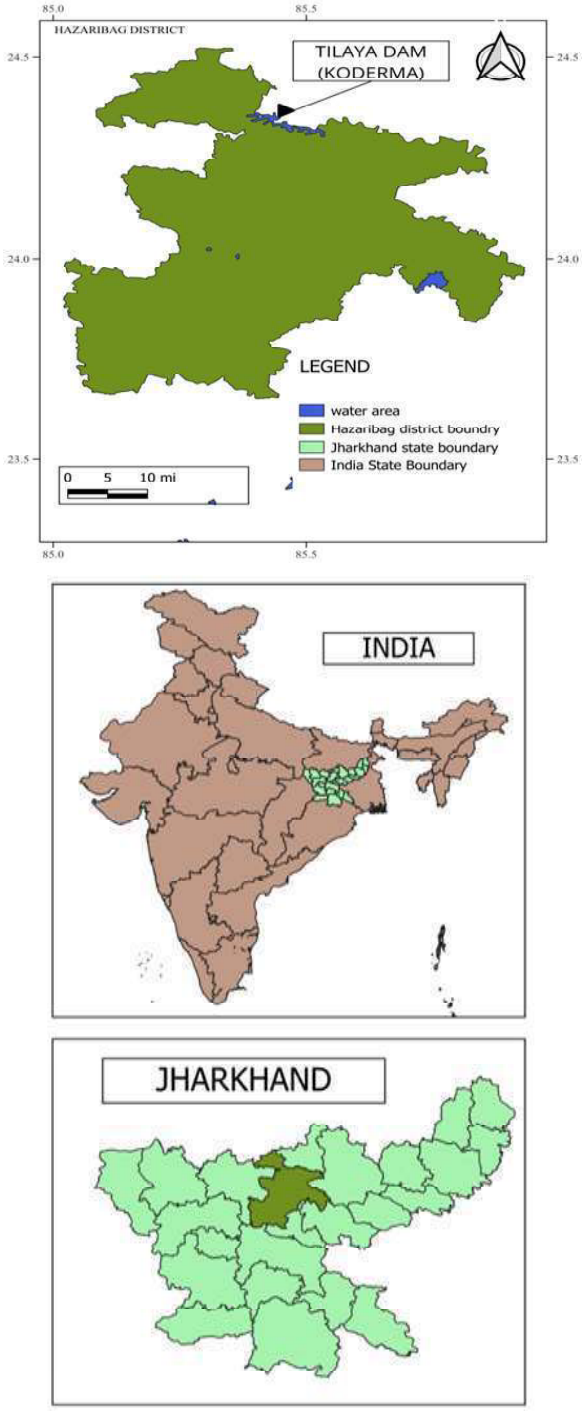


Fig.1- *Mystus bleekeri* (catchment area) collection form Tilaiya dam, Jharkhand, India

The present investigation was carried out to study the effect of hormones on the blood volume and haematocrit values in the fish, *Mystus bleekeri*, which is phylogenetically important as the fish is provided with accessory respiratory organ in the form of skin.

MATERIALS & METHODS

Large number of living specimens of *Mystus bleekeri* (Day) of approximately same size and weight (38-25 gram) were collected from the Tilaiya dam, Jharkhand (Fig:1) and were maintained in laboratory under intensive care at an average water temperature of 28°C. During captivity fishes were fed with goat liver.

For the present series of experiments, fishes were divided into three groups viz, control (Saline injected), hydrocortisone injected and L-thyroxine injected of which the latter two groups were further subdivided into three categories depending on different doses of hormones injected. Each of these categories consisted four fishes and were kept in different aquaria. The total dose of 1.0 mg, 3.0 mg. and 4.0 mg per 100 gram body weight of hydrocortisone were intraperitoneally injected in the fishes in 10 days. In another series of experiments L-thyroxine was injected intraperitoneally in total doses of 0.5 mg. 1.0 mg and 2.0 mg per 100 gram body weight in 10 days. The volume of fluid injected in different doses in the case of both the hormones remained constant. The control group of fishes also received the same sort of treatment but with sterilized Isotonic solution (NaCl)

Higher doses of Hydrocortisone acetate (4 mg/100g) and L-Thyroxine (2mg/ 100g body weight) proved lethal as there was 100% mortality.

The blood volume was determined by dye dilution technique using Evans blue (T-1824) as dye and sodium citrate as anticoagulant. At the end of the experiment the fishes were anaesthetized[8] by MS 222 (SAN 2) and were dissected to open the heart. Then known amount of Evans blue was injected into bulbus arteriosus. After 30 minutes, the concentration of dye in plasma was measured by photoelectric colorimeter. Extrapolations of the excretion slope to zero time were also calculated. Haematocrit value was determined by micro-haematocrit pipette after Van Allen. The haematocrit or packed cell volumes is a relative value of corpuscles and plasma, and from these data the total blood volume was calculated.

Table 1: Effects of different doses of hydrocortisone and L-thyroxine on the blood volumes and haematocrit values of *Mystus bleekeri* (Day) at 28 ±1°C water temperature

Sl. No	Doses of Hormones	Body Weight	Total Blood Volume			Relative % Blood Volume			Haematocrite %		
		Control & Experiments (4 animals)	Control	Experimental Value	Significance Level Value of P<	Control	Experimental Value	Significance Level Value of P<	Control	Experimental Value	Significance Level Value of P<
1	Hydrocortisone 1mg/ml/100g	25.0±0.91	0.378± 0.051	0.278±0.028	0.01	1.643± 0.221	1.108±0.113	0.01	45.0± 1.22	44.0±0.96	P<0.50
2	Hydrocortisone 3mg/ml/100g	23.0±0.59	0.378± 0.051	0.222±0.021	0.001	1.643± 0.1643	0.965±0.091	0.05	45.0± 0.45	33.0±0.84	P<0.001
3	L-thyroxine 0.5mg/ml/100g	24.0±1.47	0.051± 0.051	0.417±0.042	0.50	0.221± 0.221	1.737±0.125	0.50	1.22± 1.22	25.0±0.75	P<0.001
4	L-thyroxine 1mg/ml/100g	32.0±0.40	0.051± 0.051	0.253±0.026	0.005	0.221± 0.221	1.100±0.117	0.005	1.22± 1.22	28.0±0.71	P<0.001

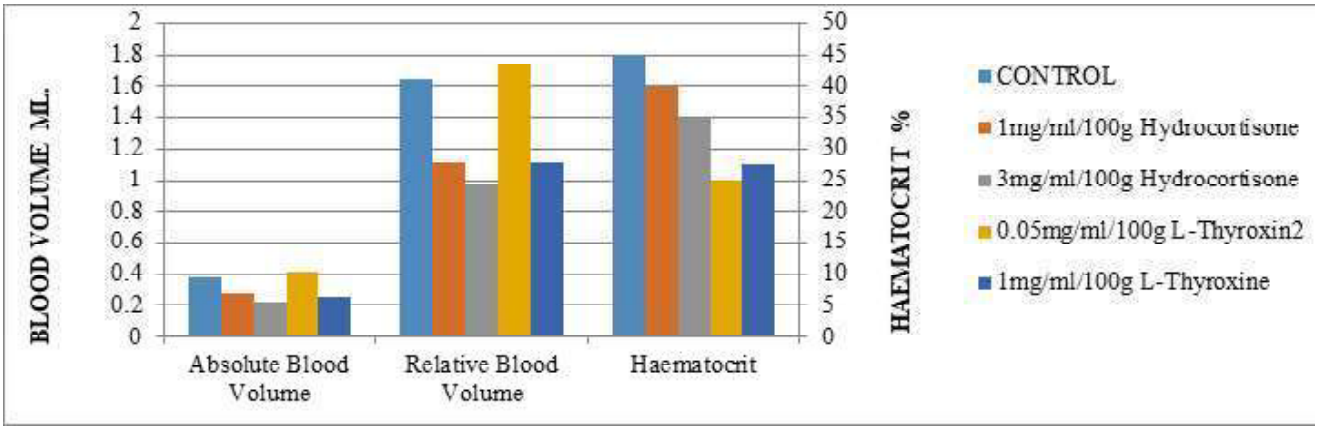


Fig. 2- Histogram showing effect of different doses of hydrocortison and L-thyroxine on blood volumes and haematocrit values

The results obtained were compared with control group and multiple t-tests were done to study the significance of their differences.

The control group had average body weight of 23.0 ± 1.58 gram with total blood volume of 0.378 ± 0.051 ml, relative (%) blood volume of 1.643 ± 0.221 ml and haematocrit value of 45.0 ± 1.22%. (Table1).

Hydrocortisone acetate treated fish with a total dose of 1 mg per 100gram body weight had total and relative body volume of 0.278 ± 0.028 ml and 1.108 ± 0.113 ml respectively. The haematocrit values were 44.0 ± 0.96% (Table 1).

Hydrocortisone acetate treated fish with a total dose of 3 mg per volume of 0.222± 0.021 ml, relative (%) blood volume of 0.965± 0.091 ml and haematocrit value of 33.0 ± 0.84 % (Table & Fig. 2).

Both the doses of hydrocortisone acetate viz, 1 mg and 3 mg/100 g. decreased the total and relative blood volume as well as haematocrit values. However, the dose of 1 mg/100 g was found to be statistically significant (P<0.01 and 0.001) so far as the total and relative blood volumes are concerned; the results of haematocrit were non-significant. But the results obtained with 3 mg/100 g hydrocortisone were quite significant (P <0.001, 0.05 and 0.001).

In L-thyroxine treated (0.5 mg/100 g body weight) fishes the total and relative blood volumes were found to be 0.417±0.042ml and 1.7373 ± 0.125ml respectively. The haematocrit values were 25.0 ± 0.75% (Table 1).

A higher dose (1 mg/100 body height) of L-thyroxine had the following effects: (1) the total blood volume of the fishes becomes 0.253±0.026 ml (2) the relative (%) blood volume achieved is 1.100 ± 0.117 ml and the haematocrit value obtained was 28.0±0.71% (Table 1)

The minimum dose of L-thyroxine (0.5 mg/100 g) however, though increases the total and relative (%) blood volumes they are non-significant. The haematocrit value decreases considerably which is found to be significant (P <0.001). The maximum viable dose of L-thyroxine (1 mg/ 100 g) decreases the total and the relative blood volume, as well as the haematocrit value and all the results were

found to be significant ($P < 0.005$, 0.005 and 0.001 respectively).

DISCUSSION

In hydrocortisone acetate treated animals the total and the relative blood volumes together with haematocrit values decreased significantly with doses of 1.0 mg and 3.0 mg per 100 g body weight but higher doses are more effective. In case of L-thyroxine treated animals the results were somewhat different in different doses. With smaller doses the total and relative blood volumes increased non-significantly but the haematocrit value decreased significantly ($P < 0.001$). Under higher doses the total, relative blood volumes and the haematocrit value were significantly decreased ($P < 0.005$, 0.005 and 0.001 respectively). From these experiments it is inferred that both the hormones reduce blood volumes and haematocrit values in this air-breathing fish.

Gabos *et al.* (1973)[9] found that in *Cyprinus carpio* L-thyroid hormone is involved in the elimination of water from the body. Martin (1969)[10] reported that the administration of thyroxine extract in normal animals accelerates all bodily process, the continuous high doses of which are fatal due to the breakdown of the body tissue caused by excessive metabolism. In this experiment also the higher doses of L-thyroxine (2.0 mg/100 g) proved to be fatal. The causes of mortality due to high doses of hydrocortisone different from that of thyroxine Hydrocortisone is a potent steroid which causes increased reabsorption of Na and CI by renal tubules, salivary glands and gastrointestinal mucosa accompanied by decline in serum K.[11] Lowering of serum K[11] by adrenal cortical steroids leads to deleterious and toxic effects upon cardiac muscles, which is followed by death from cardiac failure.[12]

The blood volume and its variations should be considered in relation to the general body fluid of the animal. Blood volume regulation is largely a question of balance between the fluid within the vessels and in the tissues. The distribution of sodium and potassium between blood plasma and tissue fluid determine the entrance or exit of water. It is probable that changes in electrolyte concentration are primary and the shift which occurs in intracellular water is secondary. The entrance or exit of water thus depends on the balance of ions inside and outside the cell. The isotonicity of the body fluid which depends mainly upon its concentration in sodium and chloride is maintained constant largely by retention or elimination of water. In this regulation kidney plays an important role.

Zarrow *et al.* (1964)[13] stated that the influence of thyroxine is reflected on protein metabolism and electrolyte balance which is seen in the deposition of intracellular mucoprotein accompanied by an increase in the intracellular level of salt and water, and modification in osmotic balance is further reflected in the blood by an increased protein content and reduced volume.

Hydrocortisone administration causes sodium and water retention which may be followed by a great increase in plasma volume and edema.[12] However, in *H. fossilis* is the effect of hydrocortisone was found to be quite different as this hormone decreases body fluid. The effects on the other hand are almost similar to the results obtained in adrenalectomized animals.[14] Gilbert (1964)[15] observed that Oestrogen, a steroid, decreases the blood volume of cockerels.

The apparent explanation of this result cannot be given with justification unless a thorough investigation on the characteristic distribution of Na^+, Cl^-, K^+ and Ca^{++} ions and the extracellular fluid volume, total tissue water and sodium space together with renal function is not determined. However, the role of adrenal in air-breathing fishes is of great physiological significance which need special attention of comparative endocrinologists and physiologists.

REFERENCES

1. **A Concise Textbook of Physiology. 1978.** India: Orient Black Swan. (https://www.google.co.in/search?hl=en&sxsrf=AJOqlzVVPs7lAp8d7ocPceJN6xatKsIItg:1675089912006&q=inauthor:%22S.+And+Kutty%22&tbm=bks)

2. **Steinbrecht K. 1957.** Narkose von Fischen. Aquarien-Terrarien-Zschr., Stuttgart (https://www.duncker-humblot.de/search/result?schnellsuche=Steinbrecht%2C+K.+%281957%29%3A+Narkose+von+Fischen.+Aquarien-Terrarien-Zschr.%2C++Stuttgart) .

3. **Thorson T. B. 1958.** Measurement of the fluid compartments of four species of marine Chondrichthyes. *Physiological Zoology*. **31(1):** 16-23. (https://doi.org/10.1086/physzool.31.1.30155374)

4. **Thorson T. B. 1959.** Partitioning of body water in sea lamprey. *Science*, **130(3367):** 99-100. (DOI: https://doi.org/10.1126/science.130.3367.99)

5. **Thorson T. B. 1961.** The partitioning of body water in Osteichthyes: phylogenetic and ecological implications in aquatic vertebrates. *The Biological Bulletin*, **120(2):** 238-254. (DOI: https://doi.org/10.2307/1539380)

6. **Prosser C. L. & Weinstein S. J. F. 1950.** Comparison of blood volume in animals with open and with closed circulatory systems. *Physiological Zoology*, **23(2):** 113-124. (DOI:https://doi.org/10.1086/physzool.23.2.30152069)

7. **Houston A. H. & DeWilde M. A. 1969.** Environmental temperature and the body fluid system of the fresh-water teleost-III. Hematology and blood volume of thermally acclimated brook trout, *Salvelinus fontinalis. Comparative Biochemistry and Physiology*, **28(2):** 877-885.(DOI: https://doi.org/10.1016/0010-406X(69)92121-5)

8. **Munshi J. D. 1961.** On the Accessory Respiratory Organs of *Heteropneustes fossilis* Bloch. *Proceedings of the Royal Society of Edinburgh, Section B: Biological Sciences*, **68(2):** 128-146. (DOI: https://doi.org/10.1017/S0080455X00000977)

9. **Gabos M., Pora E. A. & Rau L. 1973.** Effect of thyroxine (T4, TSH and thiouracil (Tu) on the oxygen consumption of the carp. *Stud. Cercet. Biol. Set. Zool.*, **25:** 39-43. (https://eurekamag.com/research/027/659/027659076.php)

10. **Martin. L. 1945.** Recent Advances in Endocrinology. Nature **156:** 34. (DOI:https://doi.org/10.1038/156034e0)

11. **White A., Handler P., Smith E. & Stetten Jr D. 1959.** Principles of biochemistry. *Principles of Biochemistry.*, (Edn 2). (https://www.cabdirect.org/cabdirect/abstract/19592703922)

12. **Best C. H. and Taylor N. B. 1967.** The physiological basis of medical practice. 8th Edition and Second Indian Reprint. The Williams and Wilkins Co., Baltimore; Scientific Book Agency, Calcutta. (https://archive.org/search.php?query=external-identifier%3A%22urn%3Aoclc%3Arecord%3A1193375854%22)

13. **M. X. Zarrow, J. M. Yochim, J. L. McCarthy and R. C. Sanborn. 1964.** Experimental endocrinology, a sourcebook of basic techniques. Pp. xvi + 519. Academic Press, New York and London. 1964. 107s. 6d. net. (DOI : https://doi.org/10.2307/4441259)

14. **Gorbman A. and Bern. H. A. 1962.** A textbook of comparative endocrinology. John Wiley & Sons, Inc., N. Y., London. (https://agris.fao.org/agris-search/search.do?recordID=US201300487037)

15. **Gilbert A. B., P. E. Lake. 1964.** The Effect on Egg Production of a Foreign Object in the Lower Oviduct Regions of the Domestic Hen, *Research in Veterinary Science*, **5(1):** 39-45, (https://doi.org/10.1016/S0034-5288(18)34823-9)